A Modification of SynthEnsemble: A Different Approach to the Fusion of Classifiers for Multi-label Chest XRay Classification

1st Reyhane Vahedi
School of Electrical and Computer Engineering,
University of Tehran,
Tehran, Iran.

2nd Behzad Moshiri
School of Electrical and Computer Engineering,
University of Tehran,
Tehran, Iran.
Department of Electrical and Computer Engineering,
University of Waterloo,
Waterloo, ON, Canada

***Abstract*—Accumulated Chest X-Ray images have widely been used in automating the diagnosis of thoracic diseases. Multiple studies on using single classifiers for this task and also some studies on the fusion of classifiers for the same purpose have been previously done. On furthering the classifier fusion approach for this problem, we have conducted a variation of the method used in the *SynthEnsemble* paper, while using the pretrained single classifiers from their work, and have reached slightly better performance results while lowering the diversity of the set of chosen classifiers and reducing the computational cost of the previously executed methods. While focusing on the concept of diversity versus accuracy for the weighted ensemble fusion, we have introduced a combination of methods for lowering the number of classifiers and have employed a few weighted ensemble methods, both from the previously mentioned paper and innovative methods of weight optimization to finally reach a [] percent improvement in multi-label image classification accuracy.**

***Index Terms*—Chest XRay, Multi-label Image Classification, Ensemble Learning, Classifier Fusion, Transfer Learning**

I. Introduction

The field of designing accurate automated diagnosis systems based on medical imaging, for various diseases and in our case and more specifically, thoracic diseases, have faced a rapidly growing surge of attention, and sequentially innovations following the growth of deep networks for classification purposes. Multiple studies have been conducted on the use of single classifiers, from classic and more basic algorithms to more advanced and complex networks, and have reached notable diagnosis accuracy. However, in the field of medical diagnoses, and because of the sensitivity of the nature of these tasks, the long-term goal is to reach 100% accuracy for conducting absolutely safe diagnoses, and thus the attempts to increase the accuracy of these systems are still made diligently and even small amounts of accuracy improvements are crucial to the flow of advancements.

On the other hand, there are multiple setbacks on the path to designing ideal automated diagnosis systems. First and foremost, there is a severe lack of detailed and structurally rich information about the types of anomalies that are detected from the chest XRay images and by doctors and radiologists of the field. And moreover, since the experts of the field rely on qualitative data to form a diagnosis from an image and there is no analytic or numeric data backing up such diagnoses, the labels given to each such image is susceptible to misdiagnosis and thus could inject an element of systematic error in the training process of each individual classifier [1].

Another issue to consider before implementing any classification or fusion method on the present data is the act that while this dataset, the Chest XRay-14 [2], is the best current dataset of chest XRay images with almost half of it being labeled, still the lack of hospital-scale labeled data is another serious hindrance in the acceleration of classification and fusion methods aiming to improve automatic diagnosis to a scale where it would be comparable to the accuracy of expert individuals in the field [2].

Notwithstanding the previous issues, there have been various utilizations of deep networks, such as Convolutional Neural Networks or CNNs, vision transformers, and models employing layers from both previous types of networks, attempting to better the circumstances of the at hand datasets and improve efficacy and accuracy, which have all shown great success in handling poorly labeled or scarce data. Furthermore, topping off the research done on single classifiers, more recent studies have tried fusing these base models in hopes of achieving a better performance accuracy. While the fusion methods vary immensely from one problem to another and depending on the chosen base models, as a general path of research, in this study, we start from simple averaging methods of decision fusion and continue towards more intricate methods, all based on different optimizations of the classifier fusion weight vector.

In this study, In comparison to the framework set by the *SynthEnsemble* [3] model of classifier fusion based on the same dataset, we have first attempted to choose a less diverse subset of classifiers and then have tried a different approach to solving the weight optimization problem, by testing the ideal weights for this more compact fusion, both under an exhaustive search

approach, and also the same global stochastic search method as the aforementioned model, or Differential Evolution [4].

II. Literature Review

Medical imaging technologies play a crucial role in modern medicine, facilitating diagnostic decisions and treatment planning. The significance of medical imaging is underscored by its consistent growth in utilization within the healthcare field. Despite the growing number of medical images, the availability of radiologists remains disproportionately low. Consequently, radiologists face an ever-increasing workload, and the cognitive burden placed on them can then possibly lead to inevitable delays in diagnosis and diagnostic errors [5]. Having in mind all that was previously said about the issues of present datasets, and the consequential influences of these issues on the aim of automating disease diagnosis, this new challenge has also added to the urgency in making efforts to integrate automated systems into the medical imaging workflow. These systems promise to enhance both efficiency and diagnostic accuracy. Notably, recent advancements in deep learning models have demonstrated diagnostic capabilities comparable to those of human experts, particularly in narrow clinical tasks across various medical domains and imaging modalities, including chest X-rays [6]. There are multiple notable instances of deep networks performing diagnosis tasks at a level comparable to that of human experts, such as but not limited to, detection of diabetic retinopathy [7], classification of skin cancer [8], identification of arrhythmia [9], and detection of pulmonary tuberculosis from chest x-rays [10].

A. *Single Classifiers*

As per previously discussed, the base classifiers used in this paper, are the same as the fusion bases used in the *SynthEnsemble* [3] model, to enable later comparison between the fusion methods of the aforementioned model and the model presented through this paper. The set of base classifiers includes two CNN models, two Vision Transformers, and two hybrid models.

*1) **CNNs**:* Convolutional Neural Networks or CNNs have emerged as a powerful tool in the field of medical image analysis, including the diagnosis of chest diseases. CNNs have demonstrated remarkable performance in automatically detecting and classifying various chest pathologies, such as pneumonia, lung cancer, and tuberculosis, from chest radiographs and other medical imaging mediums. One of the pioneering works in this area was the study by Rajpurkar et al. [11], who developed a CNN-based model called *CheXNet* that achieved radiologist-level performance in detecting 14 different chest pathologies from chest X-rays. The model was trained on our present dataset [2] and was able to outperform practicing radiologists in the detection of pneumonia. Since then, numerous studies have built upon the success of *CheXNet* and have explored various CNN architectures and training strategies for chest disease diagnosis. As for examples, we can again cite the comparative assessment of multiple pre-trained CNNs which was conducted by Wang et al. [2], who proposed a multi-label CNN model that could simultaneously detect multiple chest pathologies. Furthermore, Irvin et al. [12], developed a CNN-based model that could not only detect but also localize the specific regions of the chest affected by various diseases, by using the same dataset.

More recently, researchers have explored the use of transfer learning and ensemble models to further improve the performance of CNN-based chest disease diagnosis. Wang et al. [13] showed that by fine-tuning a pre-trained CNN model on the dataset of chest X-ray-14, they could achieve state-of-the-art results in the detection of pneumonia and other lung diseases.

Overall, the literature demonstrates the strong potential of CNN-based approaches and shows how these models have the ability to quickly and accurately detect a wide range of chest pathologies. However, In the case of using single CNN networks, further research is still needed to address challenges such as interpretability, generalization to diverse patient populations, and integration into clinical workflows.

These issues could possibly be the most impelling forces in diverting the course of research to network architectures that focus on interpetebility and generalization as well as accuracy improvement, however do not rely on the depth of a single network, or in other words, network fusion architectures. Lastly the two main CNNs used in this research are the DenseNet [14], and ConvNeXtV2 [15], which we will give a short description of in the upcoming part. However it should be noted that unlike [3], we will not be using both these networks in our final proposed architecture.

- **DenseNet**: Departing from traditional architectures, Dense Convolutional Networks (DenseNets) [14] introduce a distinctive connectivity pattern. Instead of the conventional feed-forward structure, DenseNets establish dense connections that link each layer directly to all other layers. This unique design leads to an exponential increase in the number of direct connections, mitigating the challenges associated with vanishing gradients and enabling robust feature propagation. Furthermore, DenseNets promote feature reuse and parameter reduction, which contribute to their effectiveness in optimizing image classification tasks.
- **ConvNeXt V2**: Expanding upon the foundational ConvNeXt framework, the latest iteration - ConvNeXt V2 [15] - incorporates notable architectural advancements. At its core, the model features a fully convolutional masked autoencoder structure, complemented by the introduction of a Global Response Normalization or GRN layer. The integration of these self-supervised learning techniques and architectural refinements has resulted in remarkable performance enhancements across a wide range of recognition benchmarks. Specifically performing well under training with weekly-supervised datasets such as the one used though this study. Through these innovations, ConvNeXt V2 has been setting new standards for model performance and efficiency in recent researches.

*2) **Vision Transformers**:* The emergence of Vision Transformers or ViTs has sparked significant interest in the medical

imaging community due to their impressive performance and ability to capture long-range dependencies in visual data. ViTs, inspired by the success of Transformer architectures in Natural Language Processing, have shown great promise in a variety of medical imaging tasks, including automated disease diagnosis. One of the pioneering works in this domain is the application of ViTs for COVID-19 detection from chest X-ray images. Ozturk et al. [16] proposed a ViT-based model that outperformed traditional CNNs in accurately identifying COVID-19 cases from healthy controls and patients with other respiratory illnesses. The ViT's ability to learn global and contextual features from the images proved crucial in differentiating the complex radiographic patterns associated with COVID-19, and could also be extended to resolve that these networks can also handle fine-grained features of X-ray and similar images, providing a more holistic and contextual result than previously used CNNs.

Building upon this, researchers have explored the use of ViTs for the diagnosis of other medical conditions from various imaging modalities. For instance, ViT-based models have demonstrated superior performance in detecting diabetic retinopathy from retinal fundus images [17], and classifying brain tumors from MRI scans [18]. The inherent scalability and flexibility of ViTs have allowed for their successful adaptation to diverse medical imaging tasks, often outperforming state-of-the-art CNN-based approaches. Overall, the literature showcases ViTs ability to extract meaningful features from complex medical images and provide reliable, clinically-relevant predictions. As the field continues to evolve, the integration of ViTs with cutting-edge deep learning methods is likely to drive significant advancements in computer-aided disease diagnosis, ultimately improving patient outcomes and healthcare delivery. In this study, we review two specific ViTs in preparation of choosing the best fused classifier.

- **Swin V2**: The recently introduced Swin Transformer V2 [19] presents a comprehensive set of innovative strategies that collectively drive significant improvements in the model's capabilities. At the core of these enhancements are the incorporation of a residual-post-norm architecture, the integration of cosine-based attention mechanisms, and the implementation of log-spaced position bias. Additionally, the model benefits from the incorporation of a self-supervised pre-training technique, SimMIM [20], which has been shown to enhance training stability and reduce the dependency on labeled data. The cumulative impact of these methodological advancements has been a remarkable boost in the model's performance across a diverse range of vision-centric tasks, surpassing the benchmarks set by state-of-the-art approaches [21] [22] [23]. This evolution of the *Swin* Transformer architecture underscores the continuous progress being made in the field of visual representation learning. The integration of novel training techniques and architectural refinements has enabled the *Swin V2* to push the boundaries of what is achievable in computer vision, setting new standards for model versatility and efficacy.
- **VOLO**: The Vision Outlooker or VOLO model [24] presents a novel solution to address the inherent limitations of self-attention mechanisms in computer vision tasks. At the core of this innovation is the introduction of the *outlook attention* technique, which enables the model to efficiently capture fine-grained visual features and context at a more granular level. In contrast to conventional self-attention approaches, that predominantly focus on modeling coarse-grained global dependencies, the outlook attention mechanism employed in VOLO allows for a more nuanced and localized understanding of the input visual data. This architectural difference equips the model with a enhanced ability to extract and leverage intricate low-level details, which can be crucial for accurate automated diagnosis tasks. The VOLO model's adoption of this innovative outlook attention strategy marks a significant departure from the traditional self-attention paradigm, opening up new avenues for improving the performance and versatility of vision transformers [25] [26].

*3) **Hybrid Models**:* Due to the newly discussed tremendous potential of Vision Transformers in automated disease diagnosis, and long established well performance of NNs, recent studies have investigated the integration of ViTs with other deep learning techniques or CNNs, emphasizing on the enhancement of the resulting models' self-supervised learning abilities and capabilities of multi-modal fusion, to further facilitate their diagnostic capabilities [27]. These hybrid architectures leverage the complementary strengths of different deep learning models, leading to even more accurate and robust disease detection from medical images.

- **CoAtNet**: The CoAtNets model [28] presents a unique hybrid architecture that harmoniously integrates depthwise convolution and self-attention mechanisms through the use of relative attention. This innovative approach enables a seamless fusion of convolutional and attention-based layers, resulting in enhanced generalization, capacity, and efficiency. Additionally, the systematic stacking of convolutional and attention components, designed with meticulous care, contributes to the improved performance of the CoAtNets model across various applications.
- **MaxViT**: The MaxViT model [29] introduces a novel approach that harnesses multi-axis attention to enable powerful global-local spatial interactions, even on inputs of diverse resolutions, all while maintaining linear complexity. By ingeniously combining blocked local and dilated global attention mechanisms, MaxViT seamlessly integrates attention with convolutions. This synergistic integration results in a hierarchical vision backbone, where the fundamental building block is elegantly replicated across multiple stages, enhancing the model's versatility and performance.

B. *Fusion Methods*

One particular area of focus in the field of automated disease diagnosis has been the fusion of multiple classification models to enhance the overall diagnostic accuracy and robustness. Within this domain, decision fusion has emerged as a promising approach.

Decision fusion involves the combination of predictions or decisions generated by individual classifiers to arrive at a final determination. This strategy leverages the unique strengths and perspectives of diverse models, allowing the system to capitalize on their complementary capabilities. Numerous studies have explored the application of decision fusion in medical image analysis tasks, such as the identification of diabetic retinopathy, and skin lesions which were previously discussed.

One noteworthy example is the work of Abuzaghleh et al. [30], who proposed a decision fusion framework for the automated diagnosis of skin cancer. Their approach combined the outputs of CNNs, Support Vector Machines or SVMs, and k-Nearest Neighbors or KNN classifiers using majority voting and weighted averaging techniques. The results demonstrated that the fused model outperformed the individual classifiers, highlighting the potential of decision fusion in improving the reliability and accuracy of disease diagnosis. Similarly, Alqudah et al. [31] explored the fusion of CNN-based models for the detection of COVID-19 in chest X-ray images. By leveraging decision-level fusion strategies, such as majority voting and the Dempster-Shafer theory of evidence, the authors were able to achieve significantly higher classification performance compared to standalone CNN models.

Based on all these described models, and the general framework of [3], in the Methodology section, we will get into our chosen decision fusion methods and their applicability to the problem at hand. Furthermore, in the Discussion section, we will get into the differences which have occurred between our proposed model in regards to classifiers' diversity and fusion accuracy.

C. *Dataset*

The X-Ray examination is used to diagnose the integrity of the internal organs, including the chest - heart abnormalities in size or shape or disorders of veins and arteries, lungs, pulmonary infections, tuberculosis, Pneumonia, tumors including lung cancer, and neighboring bone structures - fractures of the ribs and spine or natural distortion of the bone including the collarbone. Furthermore, it is one of the popular examinations for diagnosing lung diseases since its procedure is quite simple and can be screened initially with a low radiation amount [1]. In the context of expanding medical imaging, Wang et al. [2], introduced the ChestX-ray-14 dataset, which significantly surpasses previous datasets in terms of data volume and the percentage of labeled data. Furthermore, they conducted a comparative assessment of pre-trained Convolutional Neural Network architectures using ImageNet [32].Subsequently, researchers have explored various methodologies to enhance the detection of different chest diseases by using this specific dataset, with one notable research being done by Yao et al. [33], who first investigated the statistical dependencies among the labels of the dataset.

TABLE I
AUC OF ROC FOR SINGLE CLASSIFIERS

Model	Type	AUROC
CoAtNet	Hybrid	**0.84239**
MaxViT	Hybrid	0.84013
DenseNet121	CNN	0.82440
ConvNeXt V2	CNN	**0.84091**
Swin V2	ViT	0.83573
VOLO	ViT	0.83205

The dataset at hand, or Chest XRay-14 [2], is a comprehensive compilation of 112,120 frontal-view X-ray images was assembled from 30,805 unique patients spanning the years 1992 to 2015. This robust dataset builds upon the ChestX-ray8 collection and incorporates six additional extrathoracic conditions. One main concern surrounding this dataset is its inherent unbalancedness, meaning that more than one-third of the dataset belongs to only 3 diseases out of all 14 categories, which could cause a systematic bias of classification on the trained networks. Moreover, more than half of this dataset is unlabeled, or is labeled "No Findings", implicating the need for unsupervised learning methods.

III. METHODOLOGY

A. *Base Model Learning algorithms and Training*

Since the used dataset and base models are the same as those in [3], we only need the output of each trained model for the purpose of fusing the classifiers. Each single model has previously been trained by employing transfer learning and the re-purposing the base networks for classification of the Chest XRay-14 dataset. In each model the initial learning rate has been based on the Cyclical Learning Rate or CLR which is discussed in [34], and pretrained weights sourced from ImageNet [32] have been used for initializing each of the Neural Networks. The detailed architecture of the training processes can be found in [3]. Moreover, the output parameters and trained single models have been provided through the GitHub page of *SynthEnsemble* authors, and have been directly used in our fusion methods. As a starting point on building our fusion models, the accuracies, or the Area Under Curve(AUC) of each single model's Receiver Operating Characteristic or ROC curve, has been reported in Table. I, as per the results gained by the training and testing of the models on the Chest-X-ray-14 dataset, executed in [3]

B. *Model Exploration*

To summarize the fusion methods used in [3], after training the base models, Ashraf et al., defined two fusion approaches. One was stacking, where they concatenated the output probability or decision vectors of each individual classifier, and fed the 6x14 matrix gained per data point into a classic classifier (XGBoost), to be the final decision maker as a meta-classifier.

This method, while having worked in the literature, had the lowest accuracy of all other attempted classification methods

TABLE II
AUC OF ROC FOR STACKING, AND UNWEIGHTED AVERAGING FROM [3], AND OUR PROPOSED 4-ELEMENT UNWEIGHTED AVERAGING FUSION

Model	Stacking	6-element unweighted	**4-element unweighted**
AUROC	84.518	85.3269718	**85.4238607**

TABLE III
ACCURACY COMPARISON BETWEEN ALL OUR INNOVATIVE FUSION METHODS

Method	AUROC
4-element unweighted averaging	85.423
4-element weighted averaging + brute force optimization	87.4512
4-element weighted averaging + Differential Evolution optimization	86.1417

from this paper; even performing worse than the single classifiers CoAtNet and ConvNeXt. For this reason, we refrain from executing the same approach in the present study.

The second fusion method of [3] was Weighted Averaging. For this method, Ashraf et al. first tried a simple unweighted average of all six base classifiers, and then using that as a base line, they employed the previously discussed Stochastic Global Search method, otherwise known as Differential Evolution [4], to find the best weights for their 6-classifier weighted ensemble.

*1) **Diversity Exploration with Regards to the Average of Six Classifiers**:* While the results from [3], are remarkable in comparison to other studies done on the same dataset, and with the same classifiers in mind, there still are multiple promising fusion methods to be tested for the purpose of increasing accuracy. The present study delves into the quest of choosing more diverse and accurate base models, in order to reduce the accumulated diversity of the set of base classifiers, and thus reducing computational cost while maintaining accuracy of classification, or even improving accuracy.

For this purpose, in the present study we have employed a method of choosing the best performing model, made up of two to six classifiers. In other words, we have heuristically searched among 57 possible two to six element permutations, for the optimal unweighted model.

Through this endeavour, we have found that the accuracy of an **unweighted ensemble of the four classifiers CoAtNet, ConvNeXt, MaxViT**, and sWINv2, has a better accuracy at 85.4238%, than the unweighted average of all six classifiers, tested in [3], at an accuracy of 85.3269%. For ease of comparison, the results of the stacking fusion method, and the unweighted fusion method from [3], alongside our unweighted four-element fusion, have been reported in Table. II.

*2) **Adding Weighted Averaging Modulations to the Best Performing Permutation of Classifiers**:* As per previously discussed, the weighted ensemble used in [3] uses a Differential Evolution method for optimizing the weights of each individual classifier in the fusion. In the present study, while considering the weighted ensemble accuracy of [3], as a baseline for comparing our approaches, we concoct two different approaches to optimize the weights for our four-element weighted fusion. One of them being a brute force testing of all possible four-element arrays of weight, having in mind that the weights are bounded between 0 and 1 and also sum up to 1. And the other approach is testing the same stochastic search for optimal weights, or the Differential Evolution method, on our four-element fusion.

- **Brute Force Weight Extraction:** Having in mind that each classifier can have a weight in the range of zero to one, and that the sum of all four classifier weights should be equal to one, there are a total of 256 possible weight combinations for a four element array. By assigning these weights to the array of best classifier permutation, which included CoAtNet, ConvNeXt, MaxViT, and SwinV2, in this exact order, we can test all 256 dusion models and choose the one with the highest AUC of ROC.
- **Differential Evolution:** In this method, the same algorithm used in [3] has been employed, with the slight difference that we run this algorithm over the four chosen single classifiers and initialize each of them with their regarding weight that was calculated in the previous manner, or in other words by exhaustive search.

Lastly, by looking at Table. III, we can compare the performance of the three fusion methods proposed in this section.

IV. DISCUSSION

Thus far, we employed three different fusion methods on the general framework of the work of Ashraf et al. In both the mentioned and the present study, it is expected to see a better performance by using a more intelligent method of optimization such as Differential Evolution, however in both six-element and four-element fusions, we can see that the brute force method of assigning classifier weights performs better. Also, even though it seems that a heuristic search would be more time-consuming, but the run-time of the DE algorithm in both cases is larger than that of a global search. Accordingly, it can be concluded that using simpler methods are not necessarily a hindrance to the task of accuracy enhancement. On another note, we can decipher the results as an indication that increasing the number of single classifiers in a decision fusion architecture, however accurate they individually may be, is not necessarily an indication of a better performance of the fused classifier. By looking at Table. IV, we can see that while the brute force method has done the best out of the three fusion methods discussed in this study, the very simple unweighted averaging of four classifiers has already reached the performance of the best model of fusion presented in [3], which uses the DE algorithm.

V. CONCLUSION

Many studies underscore the value of decision fusion in automated disease diagnosis, and multiple of them were discussed through the Literature Review section.

From them we can point to a promising study done on a skin lesion analysis system for early melanoma detection,

TABLE IV
COMPARISON OF ALL FUSION METHODS[a]

Fusion Method Per Study		AUROC
The Synth Ensemble	Unweighted 6-elements	85.3269
	BF Weight Opt.	86.2252
	DE Weight Opt.	85.433
The proposed Study	Unweighted 4-elements	85.4238
	BF Weight Opt.	87.4512
	DE Weight Opt.	86.1417

[a] Opt. = optimization, DE = Differential Equation, BF = Brute Force

or another about the detection of COVID-19 from chest X-Rays, both of which have employed hybrid models in decision fusion [30] [31]. Such studies can be a positively hopeful showcasing of how the synergistic combination of multiple classifiers can lead to more robust and accurate diagnostic outcomes, gradually leading to fully automated diagnosis systems. While many studies excel in employing various new and emerging methods for enhancing the accuracy of such systems, the present study has made an attempt to implicate how taking a step back from using our computational capabilities to their utmost power, and looking towards less crowded fusion architectures that might seem less diverse, can also lead to similar or even more significant accuracy improvements. Lastly this study presents its four-element set of classifiers combined with an exhaustive optimization as a fusion method surpassing that of its predecessor model, the *SynthEnsemble* [3]. As the field continues to evolve, further exploration of advanced decision fusion techniques, including the incorporation of domain-specific knowledge and uncertainty modeling, as well as employing more intelligent or evolutionary optimization algorithms, holds the potential to drive even greater advancements in this critical area of healthcare advancement.

ACKNOWLEDGMENT

It should be noted that while being greatly inspired by the *SynthEnsemble* [3] model, as per numerously mentioned in the present work, the simulation of all the innovative fusion methods of this work was grounded upon that of the original work. The original codes can be found here and the alterations have been attached to this work, alongside a code directory describing the steps of simulation.

REFERENCES

[1] P. Rajadanuraks, S. Suranuntchai, S. Pechprasam, and T. Treebupachatsakul, "Performance comparison for different neural network architectures for chest x-ray image classification," *2021 7th International Conference on Engineering, Applied Sciences and Technology (ICEAST)*, pp. 49–53, 2021. [Online]. Available: https://api.semanticscholar.org/CorpusID:234478229

[2] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, "Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3462–3471.

[3] S. N. Ashraf, M. A. Mamun, H. M. Abdullah, and M. G. R. Alam, "Synthensemble: A fusion of cnn, vision transformer, and hybrid models for multi-label chest x-ray classification," in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 2023, pp. 1–6.

[4] R. Storn and K. V. Price, "Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces," *Journal of Global Optimization*, vol. 11, pp. 341–359, 1997. [Online]. Available: https://api.semanticscholar.org/CorpusID:5297867

[5] A. S. Hong, D. Levin, L. Parker, V. M. Rao, D. Ross-Degnan, and J. F. Wharam, "Trends in diagnostic imaging utilization among medicare and commercially insured adults from 2003 through 2016," *Radiology*, vol. 294, no. 2, pp. 342–350, 2020.

[6] S.-C. Huang, A. Pareek, M. Jensen, M. P. Lungren, S. Yeung, and A. S. Chaudhari, "Self-supervised learning for medical image classification: a systematic review and implementation guidelines," *NPJ Digital Medicine*, vol. 6, no. 1, p. 74, 2023.

[7] V. Gulshan, L. Peng, M. Coram, M. C. Stumpe, D. Wu, A. Narayanaswamy, S. Venugopalan, K. Widner, T. Madams, J. Cuadros *et al.*, "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *jama*, vol. 316, no. 22, pp. 2402–2410, 2016.

[8] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *nature*, vol. 542, no. 7639, pp. 115–118, 2017.

[9] P. Rajpurkar, A. Y. Hannun, M. Haghpanahi, C. Bourn, and A. Y. Ng, "Cardiologist-level arrhythmia detection with convolutional neural networks," *arXiv preprint arXiv:1707.01836*, 2017.

[10] P. Lakhani and B. Sundaram, "Deep learning at chest radiography: automated classification of pulmonary tuberculosis by using convolutional neural networks," *Radiology*, vol. 284, no. 2, pp. 574–582, 2017.

[11] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya *et al.*, "Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning," *arXiv preprint arXiv:1711.05225*, 2017.

[12] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya *et al.*, "Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 33, no. 01, 2019, pp. 590–597.

[13] X. Wang, X. Deng, Q. Fu, Q. Zhou, J. Feng, H. Ma, W. Liu, and C. Zheng, "A weakly-supervised framework for covid-19 classification and lesion localization from chest ct," *IEEE transactions on medical imaging*, vol. 39, no. 8, pp. 2615–2625, 2020.

[14] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 4700–4708.

[15] S. Woo, S. Debnath, R. Hu, X. Chen, Z. Liu, I. S. Kweon, and S. Xie, "Convnext v2: Co-designing and scaling convnets with masked autoencoders," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 16 133–16 142.

[16] T. Ozturk, M. Talo, E. A. Yildirim, U. B. Baloglu, O. Yildirim, and U. R. Acharya, "Automated detection of covid-19 cases using deep neural networks with x-ray images," *Computers in biology and medicine*, vol. 121, p. 103792, 2020.

[17] S. M. S. Islam, M. M. Hasan, and S. Abdullah, "Deep learning based early detection and grading of diabetic retinopathy using retinal fundus images," *arXiv preprint arXiv:1812.10595*, 2018.

[18] M. A. L. Khaniki, A. Golkarieh, and M. Manthouri, "Brain tumor classification using vision transformer with selective cross-attention mechanism and feature calibration," *arXiv preprint arXiv:2406.17670*, 2024.

[19] Z. Liu, H. Hu, Y. Lin, Z. Yao, Z. Xie, Y. Wei, J. Ning, Y. Cao, Z. Zhang, L. Dong *et al.*, "Swin transformer v2: Scaling up capacity and resolution," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 12 009–12 019.

[20] Z. Xie, Z. Zhang, Y. Cao, Y. Lin, J. Bao, Z. Yao, Q. Dai, and H. Hu, "Simmim: A simple framework for masked image modeling," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 9653–9663.

[21] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *International conference on machine learning*. PMLR, 2021, pp. 10 347–10 357.

[22] X. Chen, S. Xie, and K. He, "An empirical study of training self-supervised vision transformers," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 9640–9649.

[23] A. Brock, J. Donahue, and K. Simonyan, "Large scale gan training for high fidelity natural image synthesis," *arXiv preprint arXiv:1809.11096*, 2018.

[24] L. Yuan, Q. Hou, Z. Jiang, J. Feng, and S. Yan, "Volo: Vision outlooker for visual recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 45, no. 5, pp. 6575–6586, 2022.

[25] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 10 012–10 022.

[26] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[27] J. M. J. Valanarasu, P. Oza, I. Hacihaliloglu, and V. M. Patel, "Medical transformer: Gated axial-attention for medical image segmentation," in *Medical image computing and computer assisted intervention–MICCAI 2021: 24th international conference, Strasbourg, France, September 27–October 1, 2021, proceedings, part I 24*. Springer, 2021, pp. 36–46.

[28] Z. Dai, H. Liu, Q. V. Le, and M. Tan, "Coatnet: Marrying convolution and attention for all data sizes," *Advances in neural information processing systems*, vol. 34, pp. 3965–3977, 2021.

[29] Z. Tu, H. Talebi, H. Zhang, F. Yang, P. Milanfar, A. Bovik, and Y. Li, "Maxvit: Multi-axis vision transformer," in *European conference on computer vision*. Springer, 2022, pp. 459–479.

[30] O. Abuzaghleh, B. D. Barkana, and M. Faezipour, "Noninvasive real-time automated skin lesion analysis system for melanoma early detection and prevention," *IEEE journal of translational engineering in health and medicine*, vol. 3, pp. 1–12, 2015.

[31] A. M. Alqudah, S. Qazan, H. Alquran, I. A. Qasmieh, and A. Alqudah, "Covid-19 detection from x-ray images using different artificial intelligence hybrid models," *Jordan Journal of Electrical Engineering*, vol. 6, no. 2, pp. 168–178, 2020.

[32] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein *et al.*, "Imagenet large scale visual recognition challenge," *International journal of computer vision*, vol. 115, pp. 211–252, 2015.

[33] L. Yao, E. Poblenz, D. Dagunts, B. Covington, D. Bernard, and K. Lyman, "Learning to diagnose from scratch by exploiting dependencies among labels," *arXiv preprint arXiv:1710.10501*, 2017.

[34] L. N. Smith, "Cyclical learning rates for training neural networks," in *2017 IEEE winter conference on applications of computer vision (WACV)*. IEEE, 2017, pp. 464–472.